

Assignment 2 Submission; due Thursday, October 8, 2026

Name: Vlad Paun

Student number: 21291453

[1] Let $(\mathbf{x}_k)_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Write $\mathbf{x}_k = (x_{k1}, \dots, x_{kn})$, so that $(x_{kj})_{k \in \mathbb{N}}$ is a sequence in $\mathbb{R}$ for each $j = 1, \dots, n$.

[a] Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$. Prove that $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{a}$ if and only if $\lim_{k \rightarrow \infty} x_{kj} = a_j$ for each $j = 1, \dots, n$. (*Don't forget to prove both directions!*)

Answer:

Proof. Suppose $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{a}$. So $\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall k \geq N : \|\mathbf{x}_k - \mathbf{a}\| < \varepsilon$.

Let $\varepsilon > 0$.

$$(x_{kj} - a_j)^2 \geq 0 \quad \forall j \in [n]$$

$$(x_{kj} - a_j)^2 \leq \sum_{i=1}^n (x_{ki} - a_i)^2 \quad \forall j \in [n]$$

$$|x_{kj} - a_j| \leq \sqrt{\sum_{i=1}^n (x_{ki} - a_i)^2} = \|\mathbf{x}_k - \mathbf{a}\| < \varepsilon \quad \forall j \in [n], \forall k \geq N$$

$$|x_{kj} - a_j| < \varepsilon \quad \forall j \in [n], \forall k \geq N$$

$$\therefore \lim_{k \rightarrow \infty} x_{kj} = a_j \quad \forall j \in [n]$$

Suppose $\forall j \in [n] : \lim_{k \rightarrow \infty} x_{kj} = a_j$. So $\forall j \in [n] : \forall \varepsilon > 0 : \exists N_j \in \mathbb{N} : \forall k \geq N_j : |x_{kj} - a_j| < \varepsilon$.

Let $\varepsilon' = \frac{\varepsilon}{\sqrt{n}} > 0$. Since n is finite, take $N = \max\{N_j\}$. Then, it holds that:

$$|x_{kj} - a_j| < \frac{\varepsilon}{\sqrt{n}} \quad \forall k \geq N, \forall j \in [n]$$

$$|x_{kj} - a_j|^2 < \frac{\varepsilon^2}{n} \quad \forall k \geq N, \forall j \in [n]$$

$$\sum_{j=1}^n (x_{kj} - a_j)^2 < \varepsilon^2 \quad \forall k \geq N$$

$$\|\mathbf{x}_k - \mathbf{a}\| = \sqrt{\sum_{j=1}^n (x_{kj} - a_j)^2} < \varepsilon \quad \forall k \geq N$$

$$\therefore \lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{a}$$

□

- [b] Prove that $(\mathbf{x}_k)_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}^n$ if and only if $(x_{kj})_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$ for each $j = 1, \dots, n$. (Don't forget to prove both directions!)

Answer:

Proof. Suppose $(x_k)_{k \in \mathbb{N}}$ is Cauchy. So $\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall k, l \geq N : \|x_k - x_l\| < \varepsilon$.
Let $\varepsilon > 0$.

$$(x_{kj} - x_{lj})^2 \geq 0 \quad \forall j \in [n]$$

$$(x_{kj} - x_{lj})^2 \leq \sum_{i=1}^n (x_{ki} - x_{li})^2 \quad \forall j \in [n]$$

$$|x_{kj} - x_{lj}| \leq \sqrt{\sum_{i=1}^n (x_{ki} - x_{li})^2} = \|x_k - x_l\| < \varepsilon \quad \forall j \in [n], \forall k, l \geq N$$

$$|x_{kj} - x_{lj}| < \varepsilon \quad \forall j \in [n], \forall k, l \geq N$$

$$\therefore (x_{kj})_{k \in \mathbb{N}} \text{ is Cauchy} \quad \forall j \in [n]$$

Suppose $\forall j \in [n] : (x_{kj})_{k \in \mathbb{N}}$ is Cauchy. So $\forall j \in [n] : \forall \varepsilon > 0 : \exists N_j \in \mathbb{N} : \forall k, l \geq N_j : |x_{kj} - x_{lj}| < \varepsilon$.

Let $\varepsilon' = \frac{\varepsilon}{\sqrt{n}} > 0$. Since n is finite, take $N = \max\{N_j\}$. Then, it holds that:

$$|x_{kj} - x_{lj}| < \frac{\varepsilon}{\sqrt{n}} \quad \forall k, l \geq N, \forall j \in [n]$$

$$|x_{kj} - x_{lj}|^2 < \frac{\varepsilon^2}{n} \quad \forall k, l \geq N, \forall j \in [n]$$

$$\sum_{j=1}^n (x_{kj} - x_{lj})^2 < \varepsilon^2 \quad \forall k, l \geq N$$

$$\|x_k - x_l\| = \sqrt{\sum_{j=1}^n (x_{kj} - x_{lj})^2} < \varepsilon \quad \forall k, l \geq N$$

$$\therefore (x_k)_{k \in \mathbb{N}} \text{ is Cauchy}$$

□

- [2] The two parts of this problem are related.

- [a] Let $E \subseteq \mathbb{R}$ be nonempty. If E is bounded below, so that $\inf E$ exists, prove that there exists a sequence (a_k) with $a_k \in E$ for all k , and $\lim_{k \rightarrow \infty} a_k = \inf E$.

Answer:

Proof. Let $b = \inf E$.

From the definition of the infimum, we have that $\forall \varepsilon > 0 : \exists a_k \in E : b \leq a_k < b + \varepsilon$. Taking $\varepsilon = \frac{1}{k}$ for every k , we can construct the sequence (a_k) with $a_k \in E$ such that $b \leq a_k < b + \frac{1}{k}$. Since

$$\lim_{k \rightarrow \infty} \frac{1}{k} = 0 \quad \lim_{k \rightarrow \infty} b = b \quad \lim_{k \rightarrow \infty} b + \frac{1}{k} = b$$

The squeeze theorem gives $\lim_{k \rightarrow \infty} a_k = b$.

□

[b] Let $A \subseteq \mathbb{R}^n$ be nonempty, and $\mathbf{x} \in \mathbb{R}^n$. We define the *distance* from A to $\mathbf{x}$ to be

$$\text{dist}_A(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}.$$

(Note that $\text{dist}_A(\mathbf{x})$ is well-defined and $\text{dist}_A(\mathbf{x}) \geq 0$, because it is the infimum of a nonempty set of nonnegative real numbers.) Prove that there exists a sequence $(\mathbf{y}_k)$ in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k and $\lim_{k \rightarrow \infty} \|\mathbf{x} - \mathbf{y}_k\| = \text{dist}_A(\mathbf{x})$.

Answer:

Proof. Let $E = \{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\} \subseteq \mathbb{R}$

First, let's show that $\inf E$ exists. Since A is non-empty, then E must also be non-empty. And since all the elements of E are norms of vectors, which are always non-negative, 0 is a lower bound of E . Therefore, $\inf E$ exists. We denote $\text{dist}_A(\mathbf{x}) = \inf E$.

From [a] we get that there exists a sequence $(a_k)_{k \in \mathbb{N}}$ with $a_k \in E$ for all k and $\lim_{k \rightarrow \infty} a_k = \text{dist}_A(\mathbf{x})$.

$\forall k \in \mathbb{N}$: choose $\mathbf{y}_k \in A$ such that $\|\mathbf{x} - \mathbf{y}_k\| = a_k$, which must exist given our definition of E .

So $(\mathbf{y}_k)_{k \in \mathbb{N}}$ must be a valid sequence in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k .

Therefore, $\lim_{k \rightarrow \infty} \|\mathbf{x} - \mathbf{y}_k\| = \text{dist}_A(\mathbf{x})$. □

[3] Let $A \subseteq \mathbb{R}^n$. We define the *closure* of A , denoted by $\overline{A}$, to be the subset

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \emptyset \text{ for all } \varepsilon > 0\}.$$

That is, a point $\mathbf{x} \in \mathbb{R}^n$ lies in the closure of A if and only if every open ball of any positive radius intersects A .

[a] Show that $A \subseteq \overline{A}$.

Answer:

Proof. Let $x \in A \subseteq \mathbb{R}^n$.

$$B_\varepsilon(x) = \{y \in \mathbb{R}^n : \|x - y\| < \varepsilon\} \quad \forall \varepsilon > 0$$

By this definition, we get that $x \in B_\varepsilon(x)$ for all $\varepsilon > 0$.

Therefore, $x \in B_\varepsilon(x) \cap A \neq \emptyset$. So $x \in \overline{A}$.

Therefore, $A \subseteq \overline{A}$. □

[b] Show that $\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \emptyset \text{ for any open set } U \text{ containing } \mathbf{x}\}$.

Answer:

Proof. Let arbitrary $x \in \mathbb{R}^n$. We want to prove that $\forall \varepsilon > 0 : B_\varepsilon(x) \cap A \neq \emptyset \iff U \cap A \neq \emptyset$ for all open sets U containing x .

First, prove the forward direction:

Suppose $\forall \varepsilon > 0 : B_\varepsilon(x) \cap A \neq \emptyset$.

Take an arbitrary open set $U \subseteq \mathbb{R}^n$ containing x . Since U is open:

$$\exists \delta > 0 : B_\delta(x) \subseteq U$$

By our assumption, we know that $B_\delta(x) \cap A \neq \emptyset$. So take arbitrary $y \in B_\delta(x) \cap A$. So $y \in A$ and $y \in B_\delta(x)$. But since $B_\delta(x) \subseteq U$, we get that $y \in U$. Therefore, $y \in U \cap A$, so $U \cap A \neq \emptyset$. Since U is arbitrary, it holds for all open sets containing x .

Now, for the reverse direction:

Suppose $U \cap A \neq \emptyset$ for all open sets U containing x .

Let arbitrary $\varepsilon > 0$ and take the open set $U = B_\varepsilon(x)$ containing x . Then, we have:

$$\forall \varepsilon > 0 : B_\varepsilon(x) \cap A \neq \emptyset$$

Therefore, we have equivalence of the 2 statements. So:

$$\begin{aligned}\overline{A} &= \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \emptyset \text{ for all } \varepsilon > 0\} \iff \\ \overline{A} &= \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \emptyset \text{ for any open set } U \text{ containing } \mathbf{x}\}\end{aligned}$$

□

[c] Show that $\overline{A}$ is closed.

Hint: First show there exists ε_0 such that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \emptyset$, and then show that $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \emptyset$. You will need to use part [b].

Answer:

Proof. Let's look at $\overline{A}^c = \mathbb{R}^n \setminus \overline{A}$.

$$\forall x \in \overline{A}^c : \exists \varepsilon > 0 : B_\varepsilon(x) \cap A = \emptyset$$

Take arbitrary $x \in \overline{A}^c$. Then ε_0 exists such that $B_{\varepsilon_0}(x) \cap A = \emptyset$.

Now assume for the sake of contradiction that there exists y such that

$$y \in B_{\varepsilon_0}(x) \cap \overline{A}$$

$B_{\varepsilon_0}(x)$ is an open set containing y and $y \in \overline{A}$, so $B_{\varepsilon_0}(x) \cap A \neq \emptyset$, which is a contradiction.

Therefore, $B_{\varepsilon_0}(x) \cap \overline{A} = \emptyset$. Since x is arbitrary, it holds that

$$\begin{aligned}\forall x \in \overline{A}^c : \exists \varepsilon > 0 : B_\varepsilon(x) \cap \overline{A} &= \emptyset \\ \forall x \in \overline{A}^c : \exists \varepsilon > 0 : B_\varepsilon(x) &\subseteq (\mathbb{R}^n \setminus \overline{A}) \\ \forall x \in \overline{A}^c : \exists \varepsilon > 0 : B_\varepsilon(x) &\subseteq \overline{A}^c \\ \therefore \overline{A}^c &\text{ is open}\end{aligned}$$

Therefore, $\overline{A}$ is closed.

□

[d] If A is closed, show that $\overline{A} = A$.

Answer:

Proof. Assume A is closed. We know that $A \subseteq \overline{A}$, so we have to show $\overline{A} \subseteq A$.

Take arbitrary $y \in \overline{A}$.

$$\forall \varepsilon > 0 : B_\varepsilon(y) \cap A \neq \emptyset$$

We know A^c is open:

$$\forall x \in A^c : \exists \varepsilon > 0 : B_\varepsilon(x) \subseteq A^c$$

Which is equivalent to:

$$\forall x \in A^c : \exists \varepsilon > 0 : B_\varepsilon(x) \cap A = \emptyset$$

Now, assume for the sake of contradiction that $y \in A^c$. Since A^c is open and $y \in A^c$, there exists $\varepsilon_0 > 0$ such that

$$B_{\varepsilon_0}(y) \subseteq A^c$$

$$B_{\varepsilon_0}(y) \cap A = \emptyset.$$

But since $y \in \overline{A}$

$$B_{\varepsilon_0}(y) \cap A \neq \emptyset$$

a contradiction. Therefore,

$$y \notin A^c$$

$$y \in A$$

Since y is arbitrary, it holds for all $y \in \overline{A}$. Therefore, $\overline{A} \subseteq A$. □

(Combining parts [c] and [d] shows that A is closed if and only if $\overline{A} = A$.)

[4] Let $E \subseteq \mathbb{R}^n$ be nonempty.

[a] If $(\mathbf{x}_k)_{k \in \mathbb{N}}$ is a sequence such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$, prove that $\mathbf{p} \in \overline{E}$.

Answer:

Proof. Take arbitrary $\varepsilon > 0$.

$$B_\varepsilon(\mathbf{p}) = \{x \in \mathbb{R}^n : \|\mathbf{p} - x\| < \varepsilon\}$$

Since $\lim_{k \rightarrow \infty} x_k = \mathbf{p}$, then $\exists N \in \mathbb{N} : \forall k \geq N : \|x_k - \mathbf{p}\| < \varepsilon$

That is, $\forall k \geq N : x_k \in B_\varepsilon(\mathbf{p})$. And since $x_k \in E$ for all k , we have that

$$B_\varepsilon(\mathbf{p}) \cap E \neq \emptyset$$

Since ε was arbitrary, it holds that

$$\forall \varepsilon > 0 : B_\varepsilon(\mathbf{p}) \cap E \neq \emptyset$$

Which is exactly the condition for being a member of $\overline{E}$,

Therefore, $\mathbf{p} \in \overline{E}$. □

[b] Let $\mathbf{p} \in \overline{E}$. Prove that there exists a sequence $(\mathbf{x}_k)_{k \in \mathbb{N}}$ such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$. *Hint:* Consider $B_{\frac{1}{k}}(\mathbf{p})$.

Answer:

Proof. Since $\mathbf{p} \in \overline{E}$

$$\forall \varepsilon > 0 : B_\varepsilon(\mathbf{p}) \cap E \neq \emptyset$$

Consider the values $\varepsilon = \frac{1}{k}$ for all $k \in \mathbb{N}$. We know that for each of these values, there is at least one element both in E and $B_{\frac{1}{k}}(\mathbf{p})$. Call that element x_k .

More formally:

$$\forall k \in \mathbb{N} : x_k \in B_{\frac{1}{k}}(\mathbf{p}) \wedge x_k \in E$$

This is a sequence $(x_k)_{k \in \mathbb{N}}$ with $x_k \in E$ for every $k \in \mathbb{N}$.

Also, since $x_k \in B_{\frac{1}{k}}(\mathbf{p})$, we have that $\|x_k - \mathbf{p}\| < \frac{1}{k}$ for all k .

Take arbitrary $\varepsilon > 0$. And choose N such that $N > \frac{1}{\varepsilon}$.

$$\begin{aligned} \forall k \in \mathbb{N} : \|x_k - \mathbf{p}\| &< \frac{1}{k} \\ \forall k, l \in \mathbb{N} : \|x_{k+l} - \mathbf{p}\| &< \frac{1}{k+l} < \frac{1}{k} \\ \forall k \geq N : \|x_k - \mathbf{p}\| &< \frac{1}{k} \leq \frac{1}{N} < \varepsilon \end{aligned}$$

Since ε was arbitrary, $\lim_{k \rightarrow \infty} x_k = \mathbf{p}$. □

Taken together, these two statements say that $\overline{E}$ is exactly the set of limits of convergent sequences whose elements all lie in E .

[5] Let $(\mathbf{x}_m)_{m \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Define a new sequence $(\mathbf{s}_k)_{k \in \mathbb{N}}$ by

$$\mathbf{s}_k = \sum_{m=1}^k \mathbf{x}_m.$$

If the sequence $(\mathbf{s}_k)$ converges, then we say that the *series* $\sum_{m=1}^{\infty} \mathbf{x}_m$ *converges*. If $\lim_{k \rightarrow \infty} \mathbf{s}_k = \mathbf{s}$, then we write $\sum_{m=1}^{\infty} \mathbf{x}_m = \mathbf{s}$.

[a] Prove that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges iff: for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon$$

for all $\ell \geq k \geq N$.

Hint: A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy.

Answer:

Proof.

$$\begin{aligned} &\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall \ell \geq k \geq N : \left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon \\ \iff &\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall \ell \geq k \geq N : \left\| \sum_{m=1}^{\ell} \mathbf{x}_m - \sum_{m=1}^k \mathbf{x}_m \right\| < \varepsilon \\ \iff &\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall \ell \geq k \geq N : \|s_{\ell} - s_k\| < \varepsilon \\ \iff &(s_k)_{k \in \mathbb{N}} \text{ is Cauchy} \\ \iff &(s_k)_{k \in \mathbb{N}} \text{ converges.} \end{aligned}$$

□

- [b] Suppose that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges. For any $N \in \mathbb{N}$, prove that $\sum_{m=N}^{\infty} \mathbf{x}_m$ converges, and that

$$\lim_{N \rightarrow \infty} \left(\sum_{m=N}^{\infty} \mathbf{x}_m \right) = 0.$$

Hint: Consider the sequence $(\tilde{s}_\ell)_{\ell \geq N}$ given by $\tilde{s}_\ell = \sum_{m=N}^{\ell} \mathbf{x}_m$.

Answer:

Proof. Consider the sequence $(\tilde{s}_\ell)_{\ell \geq N}$ given by $\tilde{s}_\ell = \sum_{m=N}^{\ell} x_m$.

Since we know s_ℓ converges, let $\lim_{\ell \rightarrow \infty} s_\ell = \sum_{m=1}^{\infty} x_m = p$.

Also take arbitrary $\varepsilon > 0$.

$$\exists L \in \mathbb{N} : \forall \ell \geq L : \|s_\ell - p\| < \varepsilon$$

Now,

$$\forall \ell \geq L : \|\tilde{s}_\ell - (p - s_{N-1})\| = \|s_\ell - p\| < \varepsilon$$

Therefore, $(\tilde{s}_\ell)_{\ell \geq N}$ converges to $p - s_{N-1}$.

$$\sum_{m=N}^{\infty} x_m = p - s_{N-1}$$

Let arbitrary $\varepsilon > 0$. Since $s_k \rightarrow p$, there exists $K \in \mathbb{N}$ such that

$$\forall k \geq K : \|s_k - p\| < \varepsilon.$$

Take $N \geq K + 1$. Then $N - 1 \geq K$, and therefore

$$\left\| \sum_{m=N}^{\infty} x_m - 0 \right\| = \|p - s_{N-1}\| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary,

$$\lim_{N \rightarrow \infty} \left(\sum_{m=N}^{\infty} x_m \right) = 0.$$

□

- [6] Let $(x_k)_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Given $x \in \mathbb{R}^n$, prove that (x_k) has a *subsequence* converging to x if and only if, for all $\varepsilon > 0$ the open ball $B_\varepsilon(x)$ contains x_k for infinitely many k .

Answer:

- [7] We know that a nonconvergent sequence can have subsequences which can converge to *different* limits. Give an example of a sequence in $\mathbb{R}^n$ which has *no convergent subsequences* at all. Justify your answer. *Hint:* Recall the Bolzano-Weierstrass Theorem

Answer:

-
- [8] Recall that a subset K of $\mathbb{R}^n$ is called *compact* iff every sequence in K admits a subsequence that converges to a point in K .

Let $K_1, K_2, \dots, K_m$ be a finite collection of compact subsets of $\mathbb{R}^n$. Give two different proofs that the union $\bigcup_{j=1}^m K_j$ is compact:

- [a] using the definition of compactness.

Answer:

- [b] using the Heine-Borel theorem

Answer:

-
- [9] Let $E \subseteq \mathbb{R}^n$ be nonempty. For $x \in \mathbb{R}^n$, we define the *distance* from E to x , denoted $\text{dist}_E(x)$, to be

$$\text{dist}_E(x) = \inf\{\|x - z\| : z \in E\}$$

. Thus we can think of dist_E as a function from $\mathbb{R}^n$ to $\mathbb{R}$.

- [a] Prove that $|\text{dist}_E(x) - \text{dist}_E(y)| \leq \|x - y\|$ for all $x, y \in \mathbb{R}^n$.

Answer:

- [b] Prove that $\text{dist}_E : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}^n$.

Answer:

-
- [10] Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that f is not continuous at x for any $x \in \mathbb{R}$.

Hint: Find a way to use the sequential characterization of continuity.

Answer: